LESSON 4.3

Estimate the Percent of a Number

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can estimate the percent of a number using benchmark percents, rounding, and compatible numbers.

WHY THIS MATTERS Margaret checked the cloud storage on her phone: 6 GB total, 4 GB used. Approximately what percent of her storage is used — and of the used storage, approximately what percent are photos?

REMEMBER

Estimate the Percent of a Number

Estimating Percents of Numbers The word approximately tells you an exact answer is not required. Benchmarks, rounding and compatible numbers get you close, fast.

1. Use benchmark percents: 10% (move decimal left 1), 1% (move left 2), 50% (half), 25% (quarter).
2. Combine benchmarks (e.g. 15% = 10% + 5%; 75% = 50% + 25%).
3. Round numbers to compatible values for quick mental estimates

Words: **estimate *(estimación)*** **benchmark percent *(porcentaje de referencia)*** **compatible numbers *(números compatibles)*** **double number line *(recta numérica doble)*** **the whole *(el todo)***

PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Sort each problem by its best benchmark percent.

Groups: **Use 10% (move the decimal)** **Use 50% (take half)** **Use 25% (take a quarter)**

_____ 8.6% of 216

_____ 11% of 350

_____ 48% of 90

_____ 52% of 200

_____ 26% of 80

_____ 24% of 400

HINT Round the percent to the nearest friendly benchmark: 8.6% is close to 10%, 48% is close to 50%.

2 **Complete the table. Show how you found each value.**

Complete the benchmark table for 240.

Benchmark	Shortcut	Value
10% of 240		
1% of 240		
50% of 240		
25% of 240		

HINT 10% and 1% only move the decimal point — no long computation.

SHOW YOUR WORK

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

1 MB equals 0.001 GB. Mail uses 300 MB. About what percent of Margaret's 6 GB is mail?

(A) About 0.5%

(B) About 5%

(C) About 30%

(D) About 50%

HINT Convert first: 300 MB = ? GB.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Which fraction is equivalent to 25%?

(A) 1/4
(B) 1/25
(C) 2/5
(D) 4/1



WORK

1. **Whole** *100% is what number?*
2. **Find 10%** *Divide by 10.*
3. **Build** *Add or multiply to the percent.*
4. **Check** *Is it under or over the whole?*

5 **Circle the letter of the best answer. Show how you know.**

Write 10% as a decimal.

(A) 0.01
(B) 0.1
(C) 1.0
(D) 10.0



WORK

1. **Whole** *100% is what number?*
2. **Find 10%** *Divide by 10.*
3. **Build** *Add or multiply to the percent.*
4. **Check** *Is it under or over the whole?*

6 **Circle the letter of the best answer. Show how you know.**

Write 3/5 as a percent.

(A) 30%
(B) 35%
(C) 53%
(D) 60%

Rewrite each fraction _____

Work it out _____

Simplify _____

1. **Same size pieces** *Common denominator first.*
2. **Operate** *Only then add or subtract.*
3. **Simplify** *Divide out common factors.*

EXPLAIN YOUR THINKING

You sorted a move into 'Benchmark percent' and your partner into 'Compatible numbers.' Which number did each strategy change — the percent or the whole — and why does changing that one keep the estimate close?

SENTENCE STARTER This is ___ because it changes the ___, and the estimate stays close since ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

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Estimate the Percent of a Number

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LEARNING TARGET I can estimate the percent of a number using benchmark percents, rounding, and compatible numbers.

WHY THIS MATTERS Margaret checked the cloud storage on her phone: 6 GB total, 4 GB used. Approximately what percent of her storage is used — and of the used storage, approximately what percent are photos?

1 **Circle the letter of the best answer. Show how you know.**

Which of these has the GREATEST value: 0.7, 65%, or 3/5?

(A) 65%
(B) 0.7
(C) 3/5
(D) They are all equal

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Margaret's storage: 6 GB total, 4 GB used, and 3.5 GB of the used space is photos. The rest of the used space is mail, messages, and documents. For the question 'of the USED storage, what percent are photos,' what is the whole (100%)?

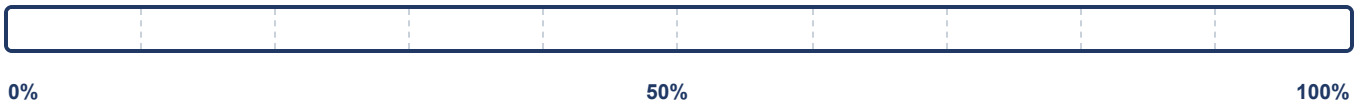
(A) 3.5 GB, the photos themselves
(B) 4 GB, the used storage
(C) 6 GB, the total storage
(D) 10 GB, adding the total and used storage together

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Margaret's storage: 6 GB total, 4 GB used, and 3.5 GB of the used space is photos. The rest of the used space is mail, messages, and documents. On a double number line built for a whole of 4 GB, what amount lines up with the 75% mark?

(A) 1 GB
(B) 3 GB
(C) 3.5 GB
(D) 4.5 GB



WORK

4 **Circle the letter of the best answer. Show how you know.**

Margaret's storage: 6 GB total, 4 GB used, and 3.5 GB of the used space is photos. The rest of the used space is mail, messages, and documents. Photos take up 3.5 of the 4 GB of used storage. About what percent of the used storage is photos?

(A) About 35%
(B) About 58%
(C) About 87%
(D) About 99%

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Estimate 19% of 412 using benchmarks or compatible numbers.

(A) About 8
(B) About 40
(C) About 80
(D) About 200

SHOW YOUR WORK

EXPLAIN YOUR THINKING

You sorted a move into 'Benchmark percent' and your partner into 'Compatible numbers.' Which number did each strategy change — the percent or the whole — and why does changing that one keep the estimate close?

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